

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓			✓

SEMESTER III

Programme	B. Sc. Statistics				
Course Code	STA3CJ201				
Course Title	Mathematical Methods for Statistics I				
Type of Course	Major				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	HSE level Mathematics course				
Course Summary Objective	Make students aware of fundamental concepts of Mathematical Analysis,				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the concepts of Real line	U	C	Instructor-created exams / Quiz
CO2	Determine limits of Sequence and series	Ap	P	Practical Assignment / Observation of Practical Skills

CO3	Understand Convergence and Divergence	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Explain Continuity and Uniform Continuity	U	C	Instructor-created exams / Home Assignments
CO5	Derivative of functions	Ap	P	One Minute Reflection Writing assignments
CO6	visualize Theory of Integration	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Real Line		16	20
	1	The Order Properties of R		
	2	Absolute Value and the Real Line		
	3	The Completeness Property of R		
	4	Archimedean Property		
	5	The Existence of $\sqrt{2}$		
	6	The Density Theorem		
	7	Nested Intervals Property		
	8	Uncountability of R		
	Sections from References:			
II	Sequences and Series		12	15
	9	Sequence, Limit of a Sequence,		
	10	Limit Theorems, Monotone Convergence Theorem (statement only), Subsequence,		
	11	Bolzano- Weierstrass Theorem		
	12	The Cauchy Criterion, Introduction to Infinite Series,		
	13	Convergence criteria, common convergence tests.		
	Sections from References:			
III	Function		10	20
	14	Limit of functions		
	15	On-sided Limits,		
	16	Continuous Functions,		